

CrAlzy MIDI: AI-powered Wearable Musical Instrumental for Novice Player

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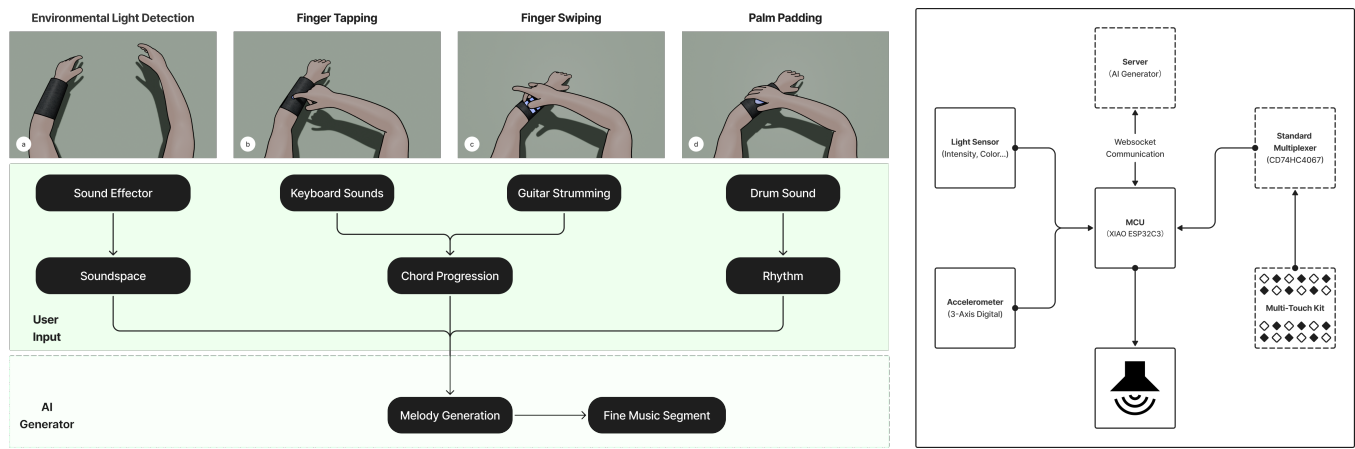


Figure 1: a) CrAlzy MIDI prototype concept: a wearable musical instrument where players control sound effects and spatial audio by adjusting environmental light color and intensity; b) Finger tapping triggers keyboard sounds and chords; c) Finger swiping simulates guitar strumming; d) Palm padding on sleeves produces drum sounds. The AI then refines the music segment based on user input, sound space, chord progression, and rhythm. Hardware principle in the right figure: The hardware design features the XIAO ESP32C3 MCU, utilizing a Multi-Touch Kit, an Accelerometer for user actions, and a light sensor for ambient light data. Communication with the AI Generator on a server is done via WebSocket. The MCU uploads raw data to the server and receives generated music data to play on the speaker. (The contest provides solid components; dotted ones are additional.)

ABSTRACT

Playing music is a deeply fulfilling and universally cherished activity, yet the steep learning curve often discourages novice amateurs. Traditional music creation demands significant time and effort to master musical theory, instrumental mechanics, motor skills, and notation reading. To lower these barriers, innovative technology-driven approaches are necessary. This proposal introduces CrAlzy MIDI, an AI-powered wearable musical instrument designed to simplify and enhance the music-playing experience for beginners. CrAlzy MIDI integrates three key technologies: wearable user interfaces, AI-generated music, and multi-modality tools. The wearable

interface allows users to play multiple instruments using intuitive finger and palm movements, reducing the complexity of traditional instruments. AI-generated music segments enable users to input a few pitches and have the AI complete the musical piece, aiding beginners in overcoming composition challenges. The multi-modality experience enhances engagement by allowing adjustments in music effects through visual stimuli such as light color and intensity changes. Together, these features make music creation more accessible and enjoyable, fostering continuous practice and exploration for novice musicians.

CCS CONCEPTS

• **Human-centered computing** → Ubiquitous and mobile devices; Interactive systems and tools; Accessibility systems and tools.

KEYWORDS

AI music, wearable user interface, accessibility

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1 INTRODUCTION

Playing music is a deeply fulfilling and universally cherished activity. However, for novice music amateurs, the learning curve can be steep and discouraging. Traditional music creation demands significant time and effort to master musical theory, understand instrumental mechanics, develop motor skills, and acquire the ability to read and interpret musical notation[9]. These challenges often deter beginners from pursuing their musical interests. To address these barriers, innovative approaches leveraging modern technology are essential. This proposal introduces CrAlzy MIDI, an AI-powered wearable musical instrument designed to simplify and enhance the music-playing experience for novices.

We propose CrAlzy MIDI, which aims to address the difficulties faced by novice musicians by integrating three key technologies: wearable user interfaces, AI-generated music, and multi-modality tools. These components collectively enhance the music-playing experience, making it more accessible and enjoyable through three key features: 1. **Wearable User Interface:** Wearable devices have been explored to create interactive musical experiences, enabling users to produce music through bodily felt experiences and movement-based interactions[1]. CrAlzy MIDI is a wearable musical instrument that allows users to play multiple instruments using their fingers and palm. This intuitive interface reduces the complexity of traditional instruments, making it easier for novices to start playing immediately; 2. **AI-Generated Music:** Previous research has shown that AI can simplify music composition by automatically generating style-matching music sections from simple inputs like chord progressions, enabling musicians of all skill levels to quickly create complex and polished demos[5]. The system utilizes AI to generate music segments. Users can input a few chords, and the AI will complete the segment, creating a coherent musical piece. This feature helps beginners overcome the challenge of composing music and allows them to focus on their creative expression; 3. **Audio-visual experience:** CrAlzy MIDI offers a rich audio-visual experience by enabling players to adjust the music effects through changes in light color and intensity. This multi-sensory approach makes the music-playing process more immersive and engaging, encouraging continued practice and exploration. Refer to Figure 1 for the interaction pipeline with CrAlzy MIDI.

Our proposed idea, CrAlzy MIDI stands out from traditional instruments by integrating AI to refine music segments based on user input and environmental variables, coupled with a wearable interface that allows intuitive control of sound effects and spatial audio through gestures and light adjustments.

2 WEARABLE USER INTERFACE

We introduced CrAlzy MIDI, a wearable user interface that enables users to play musical instrument sounds (eg., keyboard, drum, guitar) through finger and palm interactions (tapping, padding, swiping) using the Multi-Touch Kit, as shown in the right diagram

of Figure 1. This sensing scheme allows multi-touch detection on a touch-sensing matrix with a standard microcontroller, eliminating the need for specialized hardware and facilitating the capture of haptic information on wearable interfaces[8]. Similarly, Kazemitabaar et al. introduced MakerWear, a wearable construction kit for young children (K-6), which simplifies wearable creation through a tangible, 'plug-and-play' approach, providing intuitive interaction, enhanced accessibility.[7].

3 AI GENERATED MUSIC

In our proposed pipeline, we apply two stages of AI-assisted music generation. The system first helps users build raw notes and combine different mighty instrument inputs to create a melody that is sense-making and sounds good. The second stage is that AI builds upon the melody to apply different styles to the music pieces and add to other sound effects and orchestration.

Because the raw input to our system might not be perfectly aligned rhythmically and harmonically, applying adjustments which ensure the music piece maintains musical integrity and does not degrade into noise is essential in the generation process. This step ensures that the timing of notes adheres to a consistent tempo, while harmonic adjustments correct any dissonances and ensure that the chords follow a musically pleasing progression [10]. We believe these steps are crucial premises for the success of the following music style transfer stage, as the input will be modified to a clean and structured format.

In the next stage, to generate a fine music piece, a demo with only three instruments is far from satisfactory. Music style transfer is a newly-emerged area of research that focuses primarily on transforming the stylistic elements of a musical piece while preserving its original content. This process recomposes a given piece of music into a different style - such as converting classical music into jazz or transforming a piano solo into a string quartet. The potential applications of music style transfer are vast, ranging from providing therapeutic music personalized to individual preferences to creating new tools for musicians and composers to experiment with diverse styles [6]. Recent advancements in neural network architectures, such as encoder-decoder models, Variational Autoencoders (VAEs), and transformers, have significantly enhanced the ability to capture complex musical structures and dependencies, which empowers a potentially more accurate and expressive style transformation [2, 11].

4 AUDIO-VISUAL EXPERIENCE

One of the guiding principles of interactive music controller design, as proposed by Perry Cook, is to "create a complete piece of interactive music, not just an instrument or controller" [4]. Lumia, as an art form, creates a richer audio-visual experience by combining sight and sound across three dimensions: color, form, and motion. These works are dynamic and based on essentially unlimited combinations of simple elements, capable of expressing a broad range of human emotions and experiences, and can be created in real-time [3]. Our prototype extends this concept by using lighting data as a music effector, enhancing the musical experience through visual elements.

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